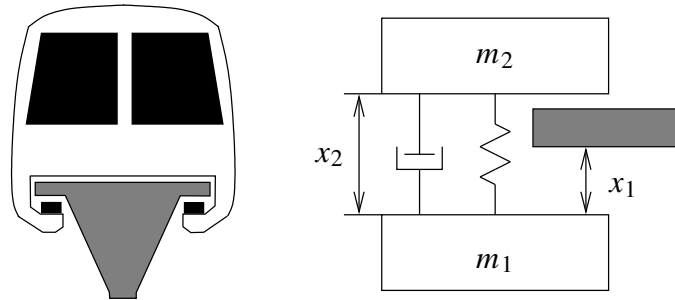


Computer Exercise: Luenberger Observer

Problem: Maglev Train

A maglev ("magnetic levitation") train is kept levitating in the air by an electromagnet that is positioned below the rail.



The system is described by the following linear model ¹:

$$\dot{x} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ -50 & 50 & -5 & 5 & -20 \\ 25 & -25 & 2 & -2 & 0 \\ -600 & 0 & 20 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -1.5 \end{bmatrix} u, \quad x(0) = \begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

where $x(t) = \begin{pmatrix} \text{distance between magnet and rail} \\ \text{distance between magnet and cabin} \\ \dot{x}_1 \\ \dot{x}_2 \\ \text{magnetic force} \end{pmatrix}$ and u is the voltage of the electromagnet.

The system is stabilised using a feedback controller with the following state feedback law:

$$u = -[775 \quad 48 \quad 19 \quad 8 \quad -21]x$$

1. Implement the closed-loop system with state feedback over a time interval $t \in [0; 5]$ in Simulink.

Hints:

- Note that an array can be specified using the format: $[a \ b; \ c \ d]$.
- Use "Gain" blocks for the matrix multiplication. In the block parameters, select "Matrix(K*u)" under "Multiplication".

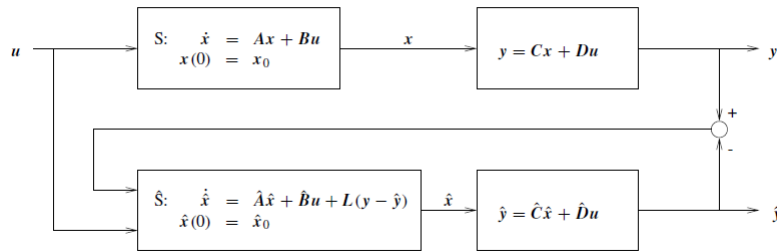
The full state feedback law cannot be used in practice since the complete state vector is not measurable, an observer is required. Assuming that only the magnetic force (x_5) can be measured:

¹J. Lunze, Regelungstechnik 2, Springer-Verlag, 2006

2. Verify the observability of the system using Matlab.
3. Use the command "place" to design an observer gain such that the eigenvalues of the estimation error system are $\tilde{\lambda}_1 = -30$, $\tilde{\lambda}_2 = -25$, $\tilde{\lambda}_3 = -20$, $\tilde{\lambda}_4 = -15$, $\tilde{\lambda}_5 = -10$.
4. Implement the observer in Simulink and replace the state feedback law ($u = -Kx$) with an output feedback law ($u = -K\hat{x}$).

Hints: Slide 3 from 2.3 Luenberger observer

General scheme:



5. Test the robustness of the observer using a disturbed system matrix for the *real* system.

$$\mathbf{A}_{disturbed} = \mathbf{A} + 0.1$$

It is assumed that the distance between magnet and rail (x_1) can be measured in addition to the magnetic force (x_5), i.e. $\mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

6. Find the observer gains and implement this system in Simulink.
7. Design and implement an LQ optimal observer with $\mathbf{Q} = \mathbf{I}$ and $\mathbf{R} = \begin{bmatrix} 10^{-4} & 0 \\ 0 & 10^{-4} \end{bmatrix}$. Use the Matlab command "lqr" for computing the observer gains.